From No Clock to Relational Time:
Page–Wooters Genesis of Clocks, AR(∞)$\to$AR(2) Decoherence, Multiplicities, and an Emergent Newtonian Limit

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Abstract

We rebuild dynamics from a "clockless" starting point using Page–Wooters (PW) conditioning. A single two-state *transfer engine* $M \in \mathrm{GL}(2,\mathbb{C})$ and an *envelope* $E = \exp\big(-s\,\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha\big)$ (s dimensionless) generate residue tracks that obey an *exact* stride-ℓ three-term identity. We show how AR(∞) parents reduce to AR(2) surrogates $(V,Q) = (2\rho\cos\theta, \rho^2)$, how decoherence appears as *phase diffusion* with rate $D = \frac{1}{2}S_\delta(0)$ after demodulation, and how multiplicities arise from residues, spin-structure, and composite monodromies. A finite-loop bridge bounds the gap between envelope transport and Levi–Civita holonomy. In a Newtonian/weak-field limit, clock-rate inhomogeneities sourced by low-frequency phase noise lead to a discrete Poisson equation whose continuum limit is $\nabla^2\Phi = 4\pi G\rho$.

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1 Symbols, Units, and Standing Conventions

Symbol	Units	Meaning
a	length	graph spacing (lattice constant)
τ	time	one update step; *per-step* quantities are dimensionless
E	—	envelope per step: $E = \exp\Big(-s\,\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha\Big)$
s	1	envelope strength $s = \sigma^2 \kappa\, \tau$ (dimensionless)
σ^2, κ	1/time, 1/length^2	envelope noise level and stiffness (appear only in s)
L_α	1/length	PSD Laplacian: $(\mathsf{I} - \alpha\mathsf{A})^\dagger(\mathsf{I} - \alpha\mathsf{A})$, choose $0 < \alpha < \lVert\mathsf{A}\rVert_2^{-1}$
$\Lambda_{\max}$	1	largest singular value of L_α (dimensionless after a absorbs units)
M	—	2×2 transfer per step ($M \in \mathrm{GL}(2, \mathbb{C})$)
$V_\ell,\ Q_\ell$	—	$V_\ell = \mathrm{tr}(M^\ell)$, $Q_\ell = (\det M)^\ell$
$r_\pm$	—	roots of $z^2 - V_\ell z + Q_\ell$, write $r_\pm = \rho e^{\pm i\theta}$ when $0 < \rho \le 1$
$\varepsilon(\mathbf{q})$	rad/step	per-step band phase; group velocity $a\nabla_{\mathbf{q}}\varepsilon$
$\Theta^a{}_i$	1	clock-tensor (coframe): $e^a = \Theta^a{}_i\, dx^i$
$\omega,\ \Omega$	1	connection one-form and curvature two-form for the induced Levi–Civita structure
$F^a{}_{ij}$	1	antisymmetric defect: $\partial_i \Theta^a{}_j - \partial_j \Theta^a{}_i$
D	1/step	dephasing rate; $D = \frac{1}{2} S_\delta(0)$ from demodulated phase noise

Group and inner product. We allow gain/loss ($M \in \mathrm{GL}(2, \mathbb{C})$). When convenient, define the SL-normalized $\widetilde{M} := M/(\det M)^{1/2} \in \mathrm{SL}(2, \mathbb{C})$. Then $\widetilde{Q}_\ell = 1$ and $\widetilde{V}_\ell = (\det M)^{-\ell/2}\, V_\ell$. Note the decimated law in (1) transforms accordingly; we do not claim V_ℓ is invariant under this normalization.

2 Layer 0: "No Clock" as Label Space with Probes and Twists

We start with a finite label set $\mathcal{E}$ (no order) and three *probes*: (i) a *localizer* that defines supports (what can co-occur), (ii) a *transfer test* that measures a 2×2 response on small supports, and (iii) a $\mathbb{Z}_2$ *twist* that flips a sign when a loop is traversed once (a Möbius identification). These are *regulators* and *diagnostics*, not ontological time. Only after choosing a section (gauge) that lists labels do we *name* indices $k \in \mathbb{Z}$ and speak about "updates." This avoids the "no-clock" paradox by keeping order as gauge, not as axiom.

3 PW Conditioning and the Exact Engine Identity

Discrete PW constraint and conditioning

Let $\mathcal{H}_C \otimes \mathcal{H}_S$ carry a joint state $|\Psi\rangle$ that satisfies a discrete PW-style constraint $(U_C \otimes U_S)\,|\Psi\rangle = |\Psi\rangle$. Conditioning on a clock sector $|t\rangle$ yields an *operational* update on S generated by a transfer M acting on two-component observables (spinors) extracted by the probes.

Universal decimated three-term law (exact)

For stride $\ell \geq 1$ and any residue track $y_t^{(r)} = x_{r+\ell t}$ obtained from a linear observable of the two-state engine,

$$y_{t+2}^{(r)} - V_\ell\, y_{t+1}^{(r)} + Q_\ell\, y_t^{(r)} = 0, \qquad V_\ell = \mathrm{tr}(M^\ell), \quad Q_\ell = (\det M)^\ell. \tag{1}$$

Cayley–Hamilton on M^ℓ

The characteristic polynomial of M^ℓ is $\lambda^2 - V_\ell\lambda + Q_\ell$. Apply it to any state segment and project with a linear functional; details as in the standard 2×2 CH proof.

Unitary specialization. If $M \in \mathrm{SU}(2)$ then $Q_\ell = 1$ and $|V_\ell| \leq 2$; write $V_\ell = 2\cos\phi_\ell$ and $r_\pm = e^{\pm i\phi_\ell}$.

4 AR(∞) Parent ⇒ AR(2) Surrogate and Decoherence

Padé reduction around a dominant conjugate pair

Assume a coarse observable has an AR(∞) representation whose transfer has a dominant pair $r_\pm = \rho e^{\pm i\theta}$ $(0 < \rho \leq 1)$. Then a [1/1] Padé match yields the AR(2) surrogate

$$x_{t+2}^{(2)} - V\, x_{t+1}^{(2)} + Q\, x_t^{(2)} = u_t, \qquad V = 2\rho\cos\theta, \quad Q = \rho^2, \tag{2}$$

which reproduces the pair exactly and pushes all other poles/zeros into the MA(∞) innovation u_t.

Model-selection diagnostics. Empirically require a gap between $|r_+|$ and the largest tail amplitude (e.g. $\gtrsim 1.2$) and whiteness of u_t (Ljung–Box at 5%).

Decoherence as phase diffusion. Demodulate the leading mode a_t and model multiplicative phase noise

$$a_{t+1} = \rho\, e^{i(\theta+\delta_t)} a_t, \qquad \mathbb{E}\,\delta_t = 0. \tag{3}$$

Then

$$|\mathcal{C}(t)| := \frac{|\mathbb{E}[a_t \overline{a_0}]|}{\mathbb{E}|a_0|^2} \;\approx\; \rho^t\, e^{-\frac{1}{2}S_\delta(0)\,t}, \qquad D := \tfrac{1}{2}S_\delta(0). \tag{4}$$

In an additive picture $a_{t+1} = r_+ a_t + \xi_t$ (small noise), $\delta_t \approx \mathrm{Im}(\xi_t/(r_+ a_t))$; estimate $S_\delta(0)$ from demodulated, amplitude-weighted residuals.

5 Birth of Clocks: Net-Zero Sources and Ring Green Functions

Consider the forced decimated law $x_{m+1} - V x_m + Q x_{m-1} = s_m$ with a net-zero probe (e.g. opposite signs half a macrocycle apart). On an L-site ring,

$$G_n^{(L)} = \frac{1}{L} \sum_{k=0}^{L-1} \frac{e^{2\pi i k n/L}}{V - 2\sqrt{Q}\cos(2\pi k/L)}.$$

A *net-zero* source has $S(0) = 0$, so the DC Fourier bin ($k = 0$) is not excited. Handle the resonance $V = 2\sqrt{Q}$ by continuity (double-root/pseudoinverse limit). For $Q = 1$, $V = 2\cos\phi$ and large L, $G_n^{(L)} \to \sin(n\phi)/\sin\phi$.

Binary spin-structure probe (PBC vs APBC). A discrete Dirac operator on a time circle has a PBC zero mode and an APBC gap π/T; in data *estimators* built from conditioned streams exhibit a sign flip $\widehat{V}_L^{\mathrm{APBC}} = -\widehat{V}_L^{\mathrm{PBC}}$ while the intrinsic $V_L = \mathrm{tr}(M^L)$ is unchanged.

Least-squares flip at stride L

$\widehat{V}_L = \dfrac{\sum_t (y_{t+2}\overline{y_{t+1}} + y_t \overline{y_{t+1}})}{\sum_t |y_{t+1}|^2}$. Under APBC, $y_t \mapsto (-1)^t y_t$: the numerator picks up (-1) and the denominator does not, hence $\widehat{V}_L \mapsto -\widehat{V}_L$.

6 Multiplicity Mechanisms and Species Labels

Residue multiplicity (stride ℓ) gives up to ℓ independent tracks (decoupled at scale set by the envelope gap). Topological/holonomy multiplicity (PBC/APBC, Möbius vs circle, $\mathrm{Pin}^{\pm}$) yields protected doublets. Composite multiplicities arise from noncommuting steps via the Magnus expansion; effective traces $V_\ell(M_{\mathrm{eff}})$ bifurcate with commutators.

Operational species label

$\Lambda = (\ell,\ V_\ell,\ Q_\ell,\ r_\pm,\ \chi \in \{\pm 1\},\ (c_0, c_1),\ \mathrm{hol})$, where χ is PBC/APBC parity, (c_0, c_1) are checksums (TKNN-like gap labels), and hol records the holonomy class.

7 Envelope Locality and Speed Limits (dimensionless s)

Per step, use $E = \exp\Big(-s\, \mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha\Big)$ with $s = \sigma^2 \kappa \tau$. A Fourier/group-velocity bound reads

$$v_{\max} \lesssim \frac{a \sup_{\mathbf{q}} \|\nabla_{\mathbf{q}} \varepsilon(\mathbf{q})\|}{1 + s\,\Lambda_{\max}^2}, \quad \text{or} \quad v_{\max} \lesssim \frac{2a\,\|\widehat{K}\|\,\Delta}{1 + s\,\Lambda_{\max}^2} \tag{5}$$

for local degree Δ. For directed/weighted graphs choose $0 < \alpha < \|\mathsf{A}\|_2^{-1}$ so that $\mathsf{L}_\alpha = (\mathsf{I} - \alpha\mathsf{A})^\dagger (\mathsf{I} - \alpha\mathsf{A})$ is well-conditioned and use singular-value radii.

8 Finite-Loop Bridge to Levi–Civita Transport

Let $e^a = \Theta^a{}_i\, dx^i$ be the coframe and $F^a{}_{ij} = \partial_i \Theta^a{}_j - \partial_j \Theta^a{}_i$ the antisymmetric defect. With mesh h and loop diameter ℓ,

$$\|P_{\mathrm{env}}(C) - H_{\mathrm{LC}}(C)\| \;\le\; C\Big(\sqrt{\int_{\Sigma(C)} \|F\|^2\, dA} \;+\; h \;+\; \ell^3 \|\nabla\Omega\|_\infty \;+\; \mu^{-1} s\, \ell\Big). \tag{6}$$

Idea of proof. (i) Strang splitting + BCH along edges bounds noncommutativity (the s-term); (ii) non-Abelian Stokes converts loop differences to surface integrals (the F-term); (iii) Poincaré/trace yields L^2 bounds on boundary integrals; (iv) Taylor remainders give the ℓ^3 term.

9 Emergent Newtonian Limit from AR(∞) Tails

In the weak-field/quasi-static regime, gravitational time dilation reads $g_{00} \approx -(1+2\Phi)$ (units $c = 1$). In our operational setting the local clock-rate shift is encoded by low-frequency phase noise $S_\delta(0, \mathbf{x})$ (after demodulation).

Discrete Poisson law (operational, calibrated)

Let the demodulated phase-increment process $\delta_t(\mathbf{x})$ be stationary with zero-frequency power $S_\delta(0, \mathbf{x})$ (units: 1/step). Define a *dimensionless* potential by

$$\Phi(\mathbf{x}) = \gamma_G\, \tau\, S_\delta(0, \mathbf{x}),$$

so that γ_G is a dimensionless calibration constant (set once by a reference redshift experiment). On a graph,

$$(\Delta_a \Phi)(\mathbf{x}) := \frac{1}{a^2} \sum_{\mathbf{y}\sim\mathbf{x}} \big(\Phi(\mathbf{y}) - \Phi(\mathbf{x})\big) = 4\pi G\, \rho(\mathbf{x}), \quad \rho(\mathbf{x}) = \alpha\, S_\delta(0, \mathbf{x}),$$

with α absorbing units (kg $\cdot$ s per step in SI). As $a \to 0$ one recovers $\nabla^2 \Phi = 4\pi G \rho$.

Assumptions and test. Quasi-static, weak-field (Newtonian gauge), and short-loop bridge error in (6). Calibrate γ_G once using a known potential difference (e.g., vertical redshift over height H), then *predict* Φ from $S_\delta(0)$ in new regions. Report $\Delta\Phi/\Phi$ and a bootstrap CI.

Sketch

The bridge (6) implies that small-loop holonomies collapse to the Levi–Civita limit controlled by F and its gradients. In the Newtonian gauge, spatial curvature is negligible and g_{00} dominates; phase slippage (clock-rate noise) accumulates along t and induces an effective potential Φ via (7). Conservation on the graph (net-zero probes) yields a discrete divergence law taking the form (8); the continuum follows from $a \to 0$.

Interpretation. AR(∞)$\to$AR(2) is a controlled coarse-graining. The coherent two-pole core is the "system"; the tail is an "environment" whose low-frequency phase power acts as a *mass density*

proxy. This aligns with gravity–decoherence hypotheses: mass-energy inhomogeneities (phase noise power) set the Newtonian potential via the Poisson law.

10 Experiments and Estimators (Reproducible Recipes)

A/B: PBC vs APBC (operational flip)

Fit AR(2) on identical streams under PBC and APBC; report $(\widehat{V}_L, \widehat{Q}_L)$ and the demodulated $(\widehat{\rho}, \widehat{D})$. Expect a sign flip of $\widehat{V}_L$; the intrinsic V_L is unchanged.

Tail and decoherence sweep

Synthetically vary tail power to scan ρ and D; compare empirical $|\mathcal{C}(t)|$ to the prediction $|\widehat{r}_+|^t e^{-\widehat{D}t}$.

Finite-loop calibration

On a grid with known curvature, vary h, ℓ, and s to measure each contribution in (6) (log–log slopes).

Minimal estimator (Python-like pseudocode).

```
# Fit AR(2): y_{t+2} = V y_{t+1} - Q y_t + u_t
# Get r_plus, demodulate, estimate D = 0.5 * S_delta(0)
```

Acknowledgments and Cross-References

The binary PBC/APBC spectral gap for the time-circle Dirac operator and the checksum demo are elaborated in our short note *"Net–Zero Yet Infinite Time"* (spectral proof, figures).

A Details: CH Proof, Ring Green's Function, and Phase-Diffusion Estimator

A. CH proof on M^ℓ

Explicit matrices, recursion for M^ℓ, and projection show (1).

B. Ring Green's function

Fourier synthesis on $\mathbb{Z}/L\mathbb{Z}$ yields $G_n^{(L)}$ and $\sum G_n^{(L)} = 0$ (net-zero).

C. Robust $S_\delta(0)$ at zero frequency

Use Bartlett weights $w_k = 1 - k/(W+1)$ on autocovariances $\gamma(k)$ up to lag W: $S_\delta(0) = \gamma(0) + 2\sum_{k=1}^{W} w_k\, \gamma(k)$, with amplitude-gated phase increments.

B Sketches: Strang+BCH, non-Abelian Stokes, and Poincaré/trace

We bound nested commutators (order s), surface terms via F, and boundary integrals by L^2 energies.

C Directed graphs: safe α and singular-value radii

Choose $\alpha \in (0, \|\mathsf{A}\|_2^{-1})$; replace eigenvalues by singular values in all spectral bounds and estimators.